



Heritage Digital Twins Beyond Modelling

Rights-Aware Architectures and Governance
Patterns for European Cultural Data
Infrastructures

EADH 2026
European Association
for Digital Humanities
Kraków

16 September 2026

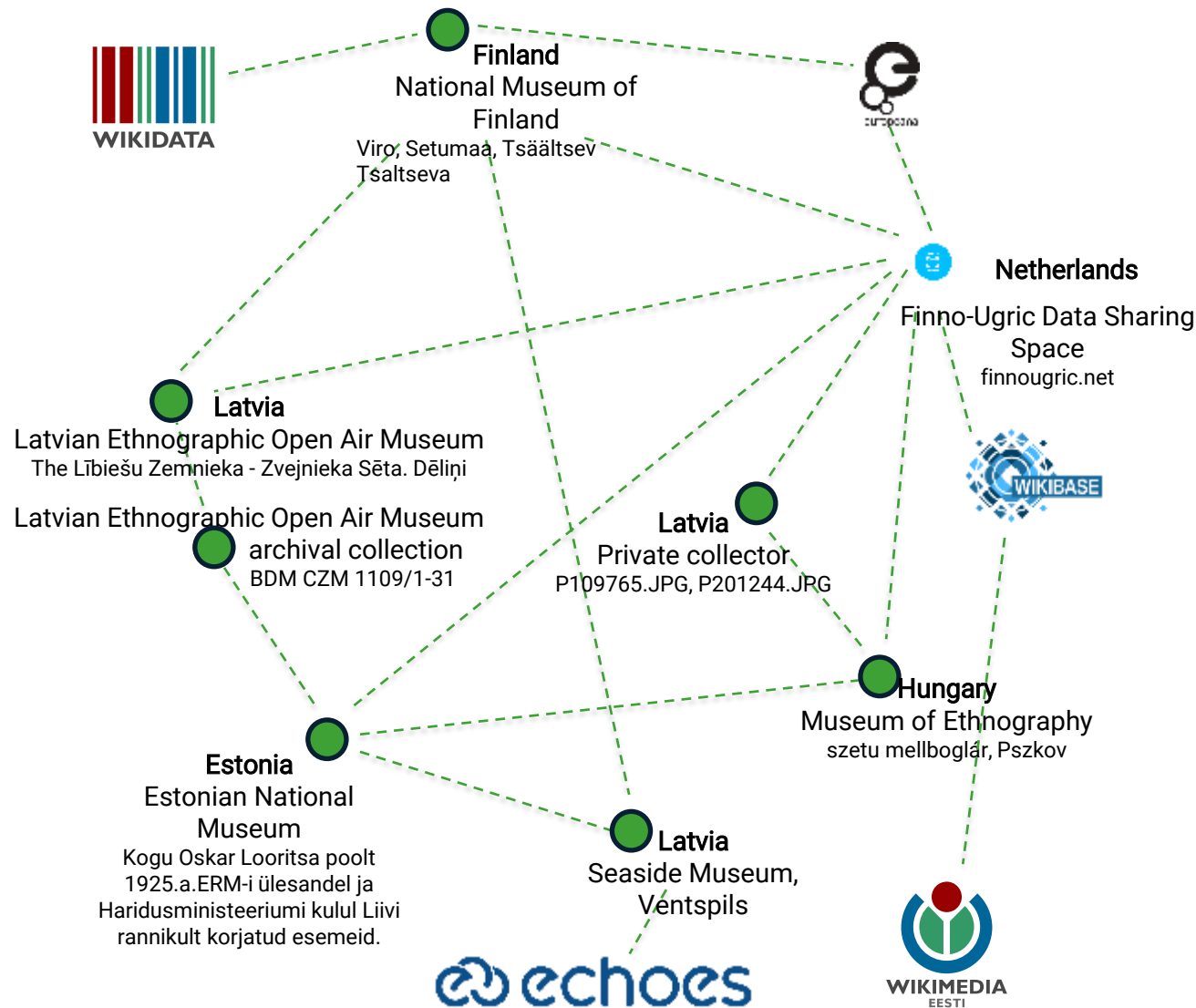
Presented by
Daniel Antal



Funded by
the European Union



Connecting dispersed knowledge



Knowledge infrastructure

Wikibase knowledge graph with Sampo-UI: widely used tools for recording and displaying graph data, with dynamic faceted search.

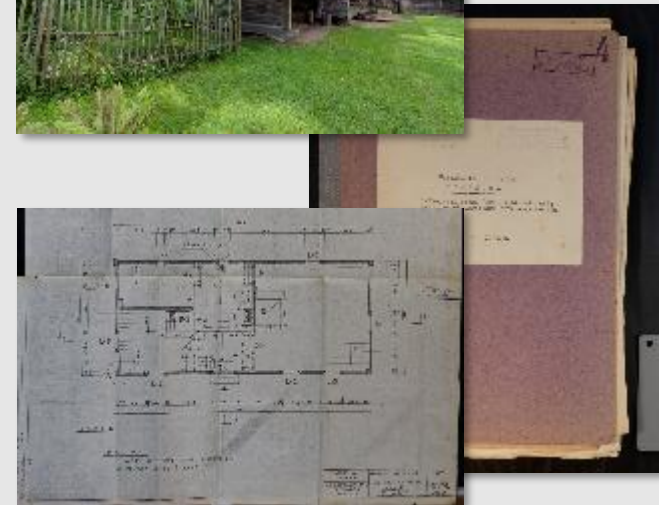
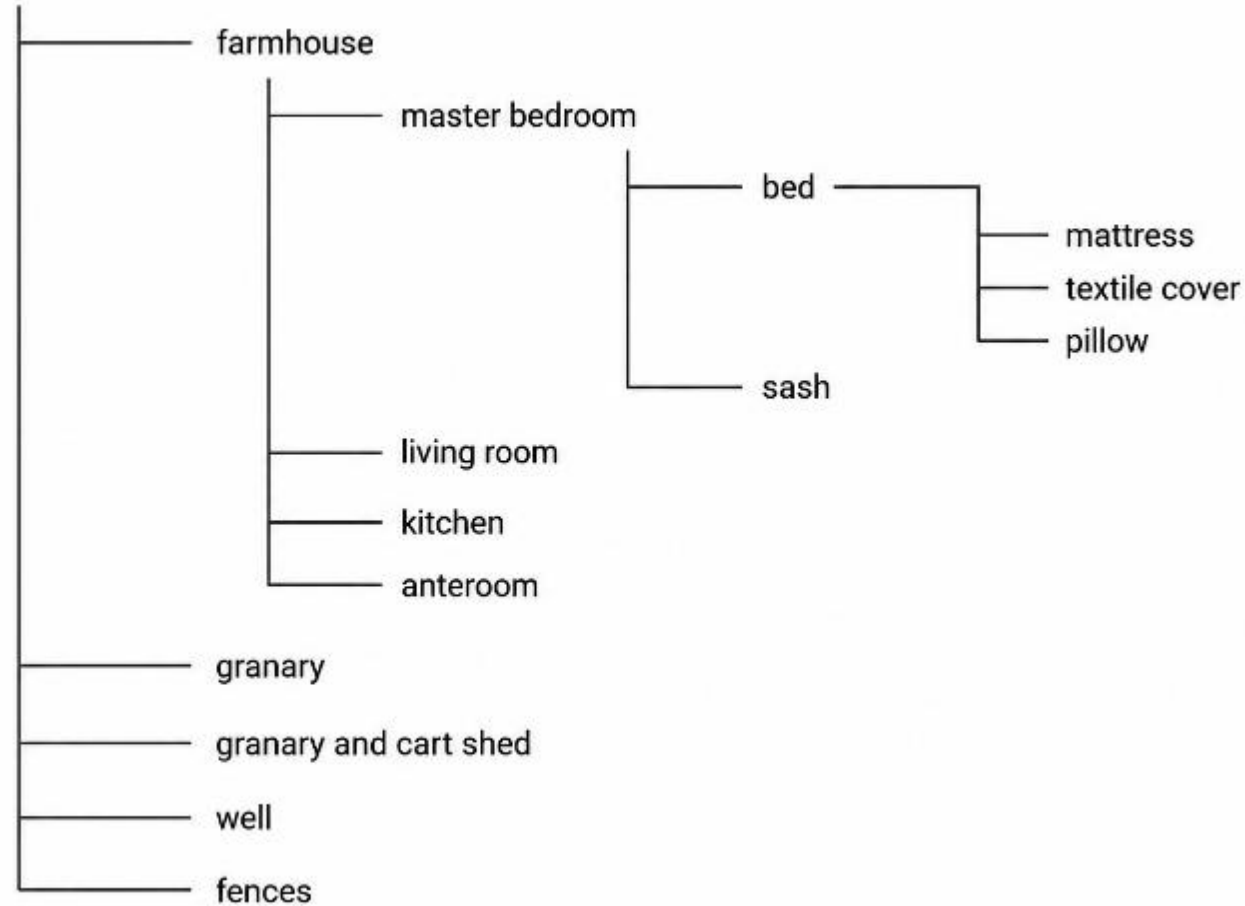
Governance infrastructure

A data sharing space enables asymmetric exchange of knowledge on an **as-needed, as-permitted** basis.

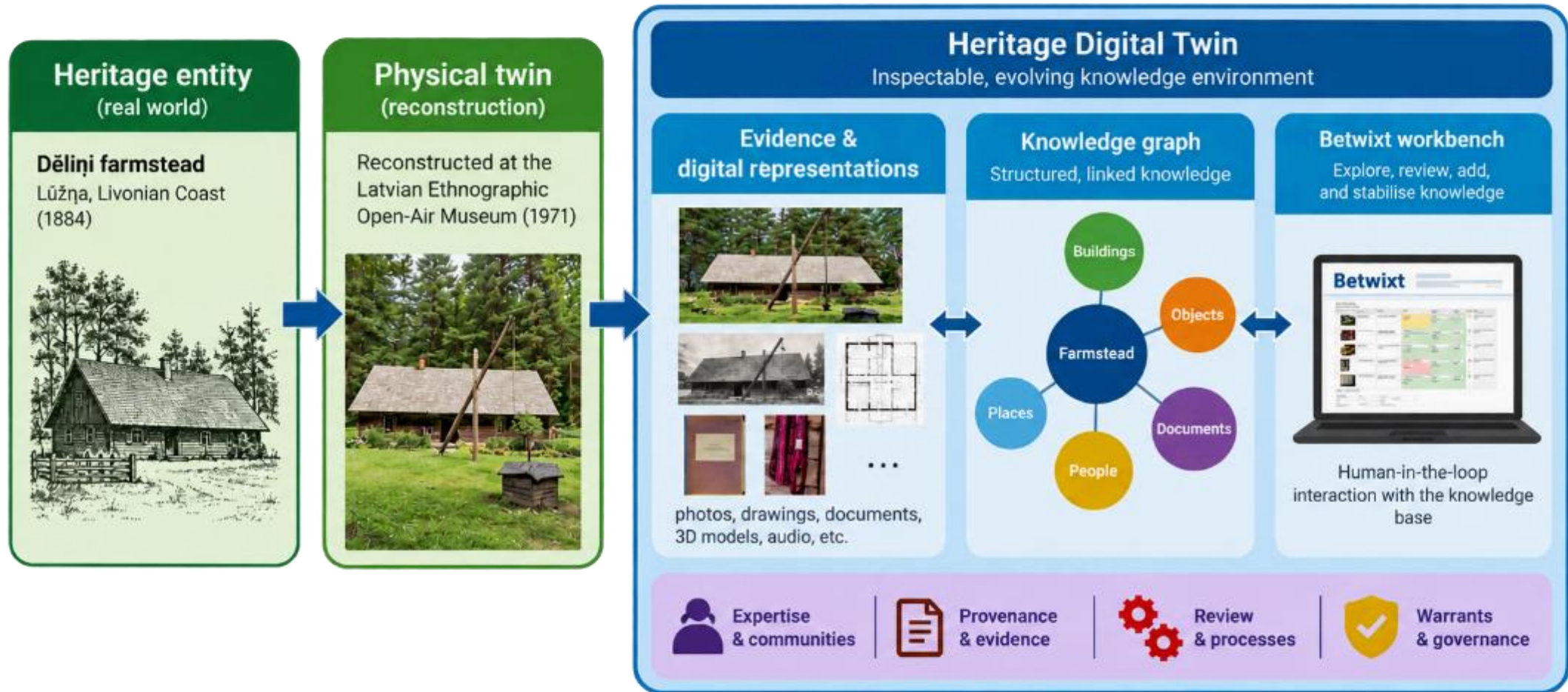
Twinning the Dēliņi farmstead

Buildings, structures, objects, archives, photographs, languages and institutions

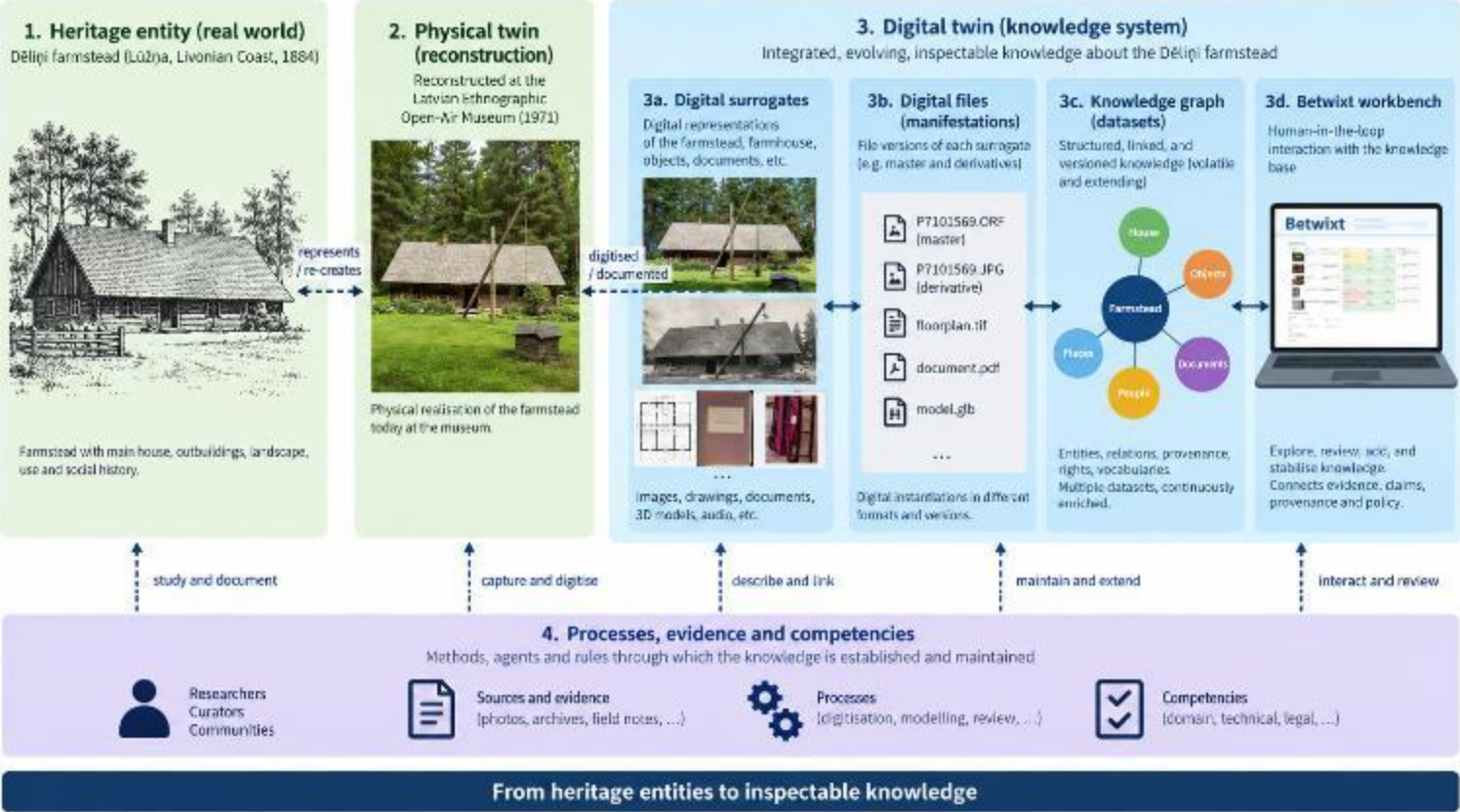
Dēliņi Farmstead



The Heritage Digital Twin is not a knowledge graph (only)



The Dēliņi Farmstead Heritage Digital Twin



Dēliņi farmstead demonstrator | REPRES 2026

Heritage Digital Twin

A maintained, inspectable state of warranted knowledge about a heritage entity, explicitly connected to the resources, assertions and processes through which that knowledge was established.

Linguistic warrant

Connects terms across languages and collections.

Evidential warrant

Records which resources or observations provide support for particular claims.

Competency warrant

Records the professional judgement needed to establish claims.

Example: competency warrant

(Human) agent(s) with the required competences review the assertion

Explicit domain knowledge	Provenance / competent assertion
The person identified in the photograph is Hilda Grīva . She died in 1982 .	Photo curator — identifies the depicted person and establishes the relevant biographical information.
The author of the photograph worked for ILFA .	Archivist — establishes the photographer’s institutional affiliation from archival evidence.
Provided that Hilda Grīva consented to the photograph and it does not depict her in a defamatory or demeaning way, her potential heirs cannot object to its use.	Legal expert — interprets the relevant legal rules and their application to the established facts.

Hilda Grīva



Born

Hilda Cerbach
February 27, 1910
Sikrags, Courland Governorate,
Russian Empire

Died

September 13, 1984 (aged 74)
Ventspils, Courland, Latvian
SSR

Burial place

Sikrags cemetery

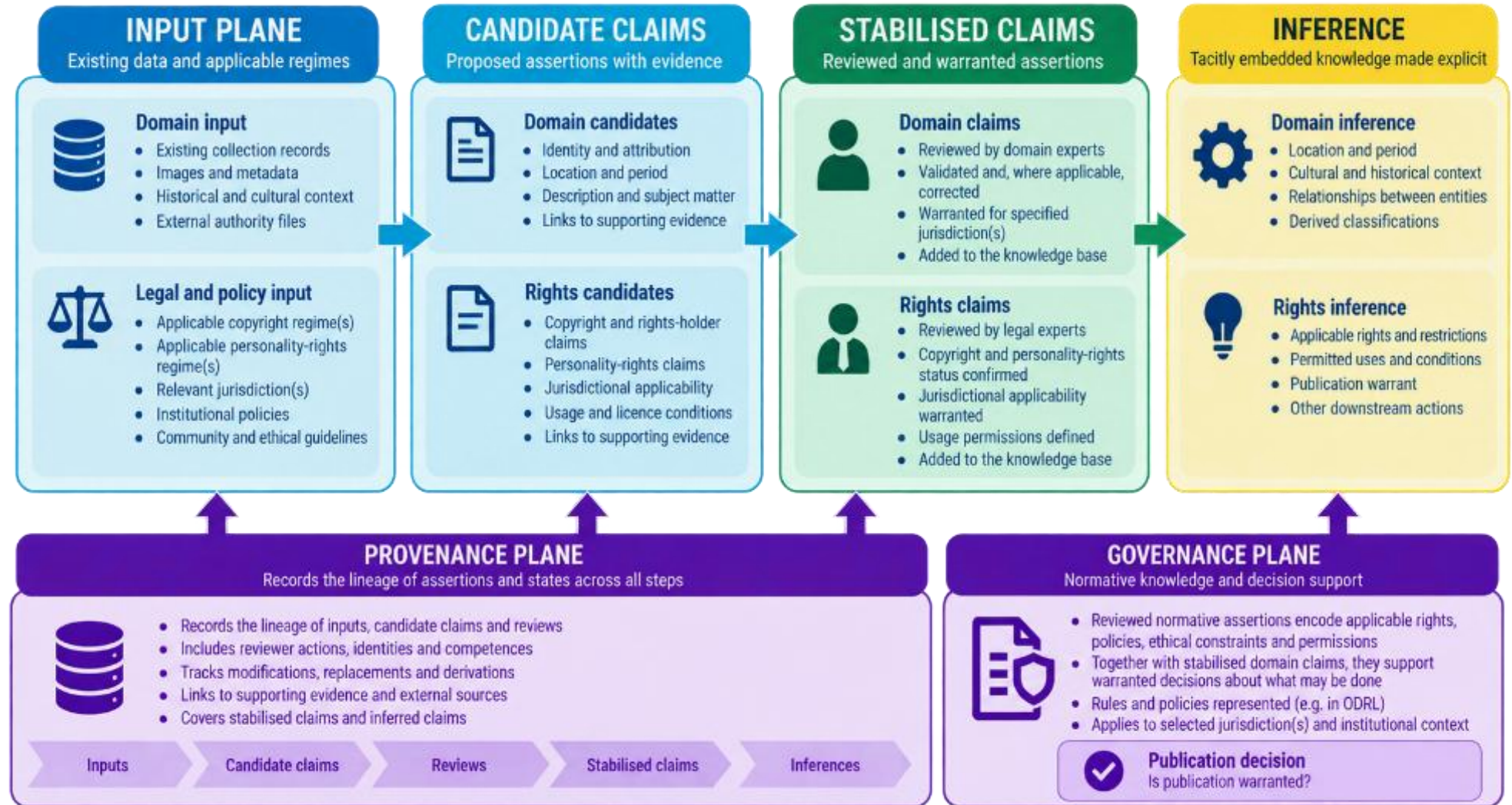
Occupations

Musician, music teacher

Known for

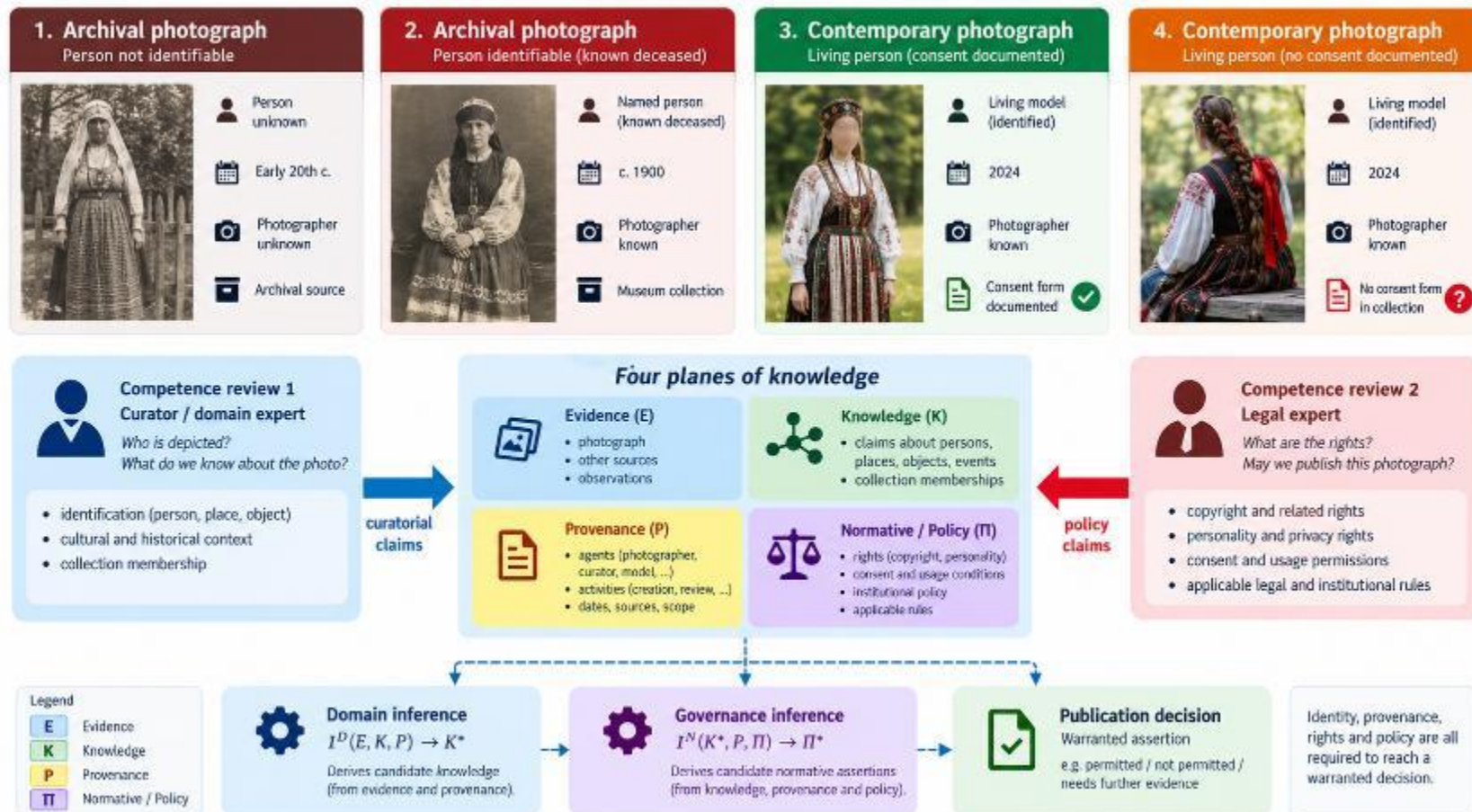
Founder and artistic director of
the Livonian folklore ensemble
"Kāndla"

Four dimensions of warranted knowledge



Can we publish this photograph?

Looking into Europe's digital black hole



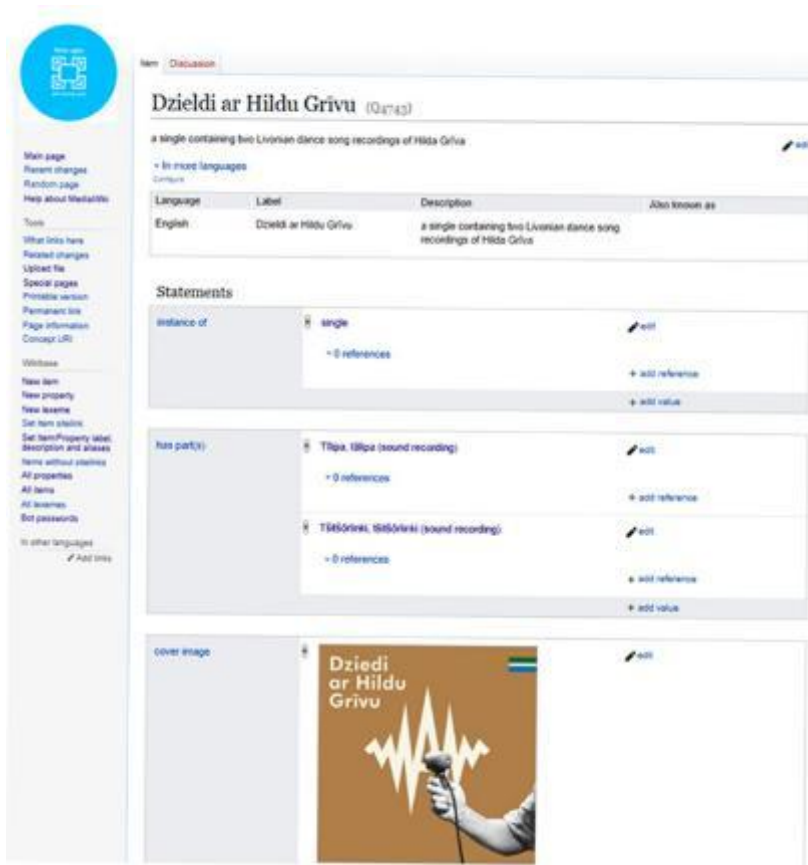
Digitised ≠ usable

A Heritage Digital Twin may be able to **play a Livonian folk song**, yet lack permission to make the mechanical copy needed for **MIR or AI analysis**—or conversely, permit computational research while prohibiting publication to a wider audience.

Large parts of Europe's audiovisual heritage are already digitised but remain unavailable online. Even among European films produced between **1996 and 2020, only about 59% were available on VOD**; availability is substantially lower for older works.

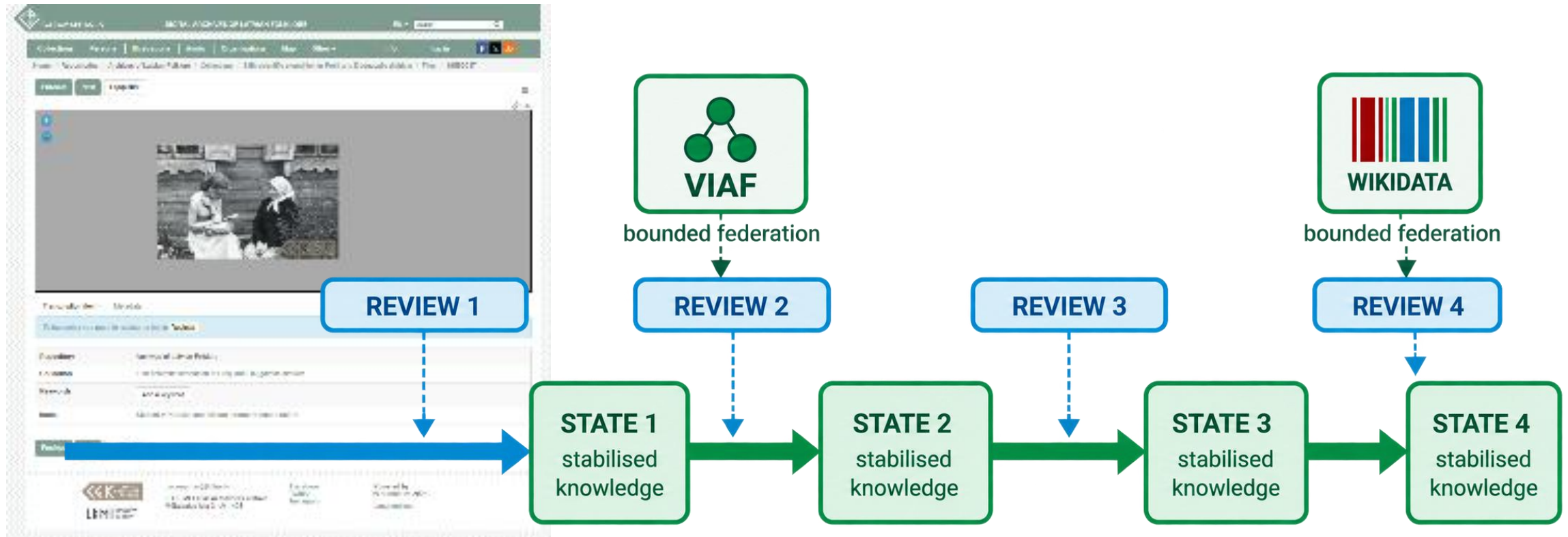
From archival object to public communication

Rights-aware state transitions can make heritage usable in new contexts and applications



From one warranted state to the next

Review and bounded federation create an inspectable path of semantic knowledge production.



Betwixt: a workbench for inspection

Exercise 3: From wide data to atomic assertions

Review the proposed classifications and heritage relationships. Information shown as context may help you make a decision, but contextual information is not itself an assertion under review.

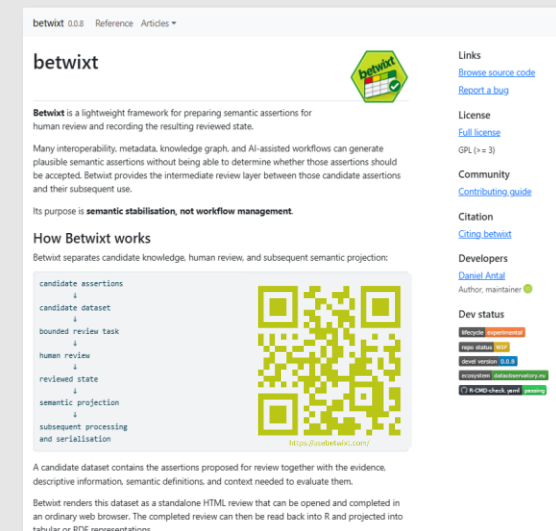
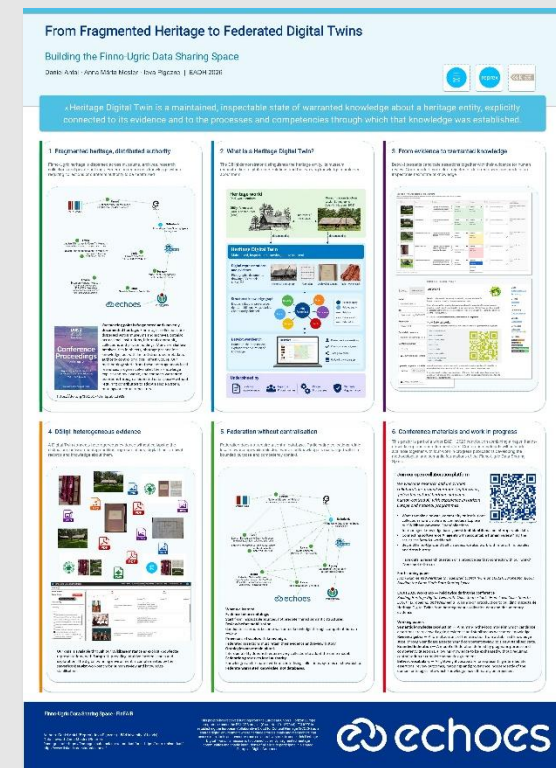
#	Evidence	Label	Description	subject	instance_of	heritage_of	Finalise	Reviewer comment	context_held_by Context — not reviewed
1	 If you are unsure, you can visit the link	Dēliņi farmstead	a farmstead, constructed between 1819 and the early 20th century, form a distinctive three-courtyard layout, originally built in the village of Lūžna	Dēliņi farmstead Definition ↗ Defer Reject	farmstead Definition ↗ Defer Reject	Livonians Definition ↗ Defer Reject	☐	Optional comment on row 	Ethnographic Open-Air Museum of Latvia
6	 If you are unsure, you can visit the link	Dēliņi farmhouse	the main housing unit of the Dēliņi homestead	Dēliņi farmhouse Definition ↗ Defer Reject	building Definition ↗ Defer Reject	Livonians Definition ↗ Defer Reject	☐	Optional comment on row 	Ethnographic Open-Air Museum of Latvia
8	 If you are unsure, you can visit the link	Dēliņi farmstead: granary (BDM)	The granary building facing the main housing unit of the Dēliņi homestead (BDM 8071)	Dēliņi farmstead: granary (BDM 8071) Definition ↗ Defer Reject	building Definition ↗ Defer Reject	Livonians Definition ↗ Defer Reject	☐	Optional comment on row 	Ethnographic Open-Air Museum of Latvia
11	 If you are unsure, you can visit the link	sash (BDM zpn 8246)	The sash (BDM zpn 8246) is a black, pink and red tablet-woven sash which, according to the list of inventory, was used as reins	sash (BDM zpn 8246) Definition ↗ Defer Reject	sash Definition ↗ Defer Reject	Latvians Definition ↗ Defer Reject	☐	Optional comment on row 	Ethnographic Open-Air Museum of Latvia
20	 If you are unsure, you can visit the link	Dēliņi combined cowshed-barn	a combined cowshed and barn built originally around 1870 in Lūžna, on the Livonian coast. It was moved to the Ethnographic Open-Air Museum	Dēliņi combined cowshed-barn (BDM 8069) Definition ↗ Defer Reject	building Definition ↗ Defer Reject	Estonians Definition ↗ Defer Reject	☐	Optional comment on row 	Ethnographic Open-Air Museum of Latvia

0 finalised · 0 deferred cells · 0 rejected cells

Heritage Digital Twins as knowledge production

Preserve not only what we know, but how knowledge is established, governed and changed.

1. Creating the twin was the easier part — more research effort went into **governance** than into digital representation.
2. Humans, software agents and AI agents can investigate the same heritage resources and contribute candidate knowledge. Rights-awareness must be built in, not added afterwards.
3. Digital Humanities must **rethink governance for computational research** — not only copyright and personality rights, but CARE, scientific integrity, competence, provenance and accountability.
4. The Dēliņi twin is a test case for **human-centred computation and trustworthy AI** — enabling experimentation while keeping claims inspectable, actions governed and responsibilities attributable.
5. We have a **poster** that invites you to try out **our open-source tools**, and we held yesterday two workshops. Those materials will be made available on <https://reprex.nl/observatories/finno-ugric/> and our <https://finnougric.substack.com> blog.



Thank you for your attention!

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